

Amudha Balamurugan

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#) • [Website](#)

Cloud Engineer | AWS | CI/CD Automation | Cloud Security

Cloud Engineer with hands-on experience architecting, deploying, and optimizing AWS and hybrid cloud environments. Skilled in cloud security, application migration, infrastructure automation, and proactive system monitoring. Passionate about delivering resilient, scalable, and secure solutions for enterprise environments. Committed to supporting mission-driven organizations with operational excellence and innovation.

Core Competencies

- **Cloud Engineering:** AWS (EC2, Lambda, S3, IAM, Fargate), GCloud (basic)
- **Containerization:** Docker, ECS, Fargate; CI/CD integration
- **Infrastructure Automation:** Terraform, CloudFormation, Ansible
- **Monitoring & Troubleshooting:** CloudWatch, Splunk, proactive root cause analysis
- **Security & Compliance:** IAM, VPC Security Groups, Security+, Encryption Standards
- **DevOps & CI/CD:** GitHub Actions, Jenkins, Git, YAML pipelines
- **Migration & Optimization:** Application modernization, IaaS/PaaS cloud enablement
- **Documentation & Collaboration:** Lucidchart, Confluence, Agile/SCRUM teamwork

Professional Experience

Cloud Engineer – Freelancer | Immigrant Buddies (Remote, Feb 2025 – Present)

- Designed and provisioned secure AWS cloud environments using **Terraform** for high-availability, scalable solutions
- Led the containerization of workloads using **Docker** and **Fargate**, improving deployment speed and portability
- Developed and managed **CI/CD pipelines** with GitHub Actions to automate testing, release, and rollback workflows
- Implemented CloudWatch monitoring, established proactive alerting, and conducted regular root cause analysis to prevent outages
- Authored and maintained system documentation, deployment runbooks, and operational support guides

DevOps Engineer Trainee | Level Up in Tech (Remote, Jan 2024 – Jan 2025)

- Built cloud-native applications leveraging AWS services, including Lambda, API Gateway, DynamoDB, and EC2
- Participated in cloud migration projects, configuring IaaS/PaaS resources, user access, and environment optimization
- Supported security audits by enforcing cloud security best practices, identity management, and encryption at rest/in transit
- Gained hands-on exposure to Kubernetes cluster deployments and container orchestration for hybrid cloud use cases
- Communication strategies in mock client scenarios, including whiteboarding and requirement walkthroughs

Junior Web Developer | Cloud City Partner (Remote, Oct 2022 – Jun 2023)

- Supported cloud-hosted web applications, DNS routing, SSL/TLS implementation, and server configuration tuning
- Conducted performance monitoring, error logging, and patching operations across multiple cloud environments
- Worked in Agile sprint cycles and participated in retrospectives and technical reviews

Key Projects

Collect, process, and store data in BigQuery Dec 2024

TheLook Fintech

- Developed and implemented a plan to help the Treasury Department use data effectively to track its performance and growth
- Developed a comprehensive data analysis project leveraging Google BigQuery for efficient querying and data aggregation from a large dataset.
- Improved data accessibility for non-technical stakeholders, resulting in faster decision-making and a clearer understanding of data-driven opportunities.
- Analyzed a company's historical sales data using Python and SQL to identify trends, high-performing products, and seasonal patterns.

Optimizing Deployments with EC2, S3, IAM, and Jenkins Integration using Terraform May 2024 - Jun 2024

Level Up in Tech

- Provisioned EC2 with Ubuntu, integrated Jenkins for CI/CD, and configured S3 for artifact storage, achieving a 30% reduction in deployment time.
- Configured IAM roles and policies to strengthen security, following best practices in access control and compliance.
- Integrated Jenkins with AWS services, automating the build, test, and deployment processes, resulting in a 30% reduction in deployment time and improved team collaboration.

AWS Fargate-Based Microservice Deployment

- Containerized backend services and deployed them via **Fargate**, enabling cost-effective autoscaling without EC2 management

Cloud Monitoring and Proactive Incident Response

- Implemented proactive monitoring with **CloudWatch**, enabling early detection of performance degradation and rapid incident response

Infrastructure as Code Automation

- Created a fully automated, repeatable infrastructure provisioning using **Terraform** and GitOps practices

Certifications

- **AWS Certified Solution Architect Associate**– 03/2025–03/2028
- **AWS Certified AI Practitioner**– 03/2025–03/2028
- **AWS Certified Developer Associate** – 04/2024–04/2027
- **AWS Certified Cloud Practitioner** – 05/2022–05/2028
- **Terraform Associate** – 09/2024–09/2027
- **Splunk Core User** – 11/2023–11/2026
- **Google Cloud Cybersecurity**
- **Google Data Analytics**

Education

AWS Cloud DevOps Engineer, Level Up in Tech (2024)

AWS Re/Start Graduate, Per Scholas (2022)

M.Tech in Technology, National Institute of Technology (India, 2005)

B.E. in Engineering, University of Madras (India, 1995)

Additional Qualifications

- U.S. Citizen; eligible for DoD background check and PIV credentialing
- Strong interpersonal skills; proven success working with cross-functional and security-focused teams
- Rapid learner, committed to staying current with cloud technologies and cybersecurity standards